

Week 10 — Activity: Solving the Heat Equation as a Human Computer

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To develop intuition for finite difference methods by acting as the nodes in a numerical heat flow model.

In a numerical model, each point (node) updates its temperature using only information from its neighbours.

In this activity, you will act as the nodes in a finite difference model.

Box .1: Human Computer

Before electronic computers existed, a *computer* was literally a person employed to carry out calculations by hand or with mechanical calculators. Large groups of human computers were used in astronomy, navigation, ballistics, engineering, and later for the numerical solution of differential equations. By the late 19th and early 20th centuries, many of these computing groups were composed largely of women. Examples include the astronomical computers working for the *Nautical Almanac* and later teams at observatories, government laboratories, and research institutions.

During World War II, the scale of numerical work *exploded* (pun intended). Hundreds of human computers computed ballistic firing tables and solved systems of finite-difference equations for artillery trajectories, fluid flow, and other military problems. Post-WWII, human computers were also instrumental to the US space program. If you've seen the movie *Hidden Figures*, you learned about some of these remarkable women who helped make going to the Moon possible by producing many of the calculations needed to determine flight plans, rocket burn parameters, launch windows, and safe re-entry trajectories.

Finite-difference methods were especially important because many physical problems could not be solved analytically. Human computers discretized differential equations onto grids and repeatedly computed numerical updates row by row—an extremely laborious process that helped motivate the development of electronic computers. The first six programmers of the world's first general-purpose electronic computer, ENIAC, were women who had previously worked as human computers.

David Alan Grier, **2005**, *When Computers Were Human*, Princeton University Press.

Kathy Kleiman, **2022**, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer*.

David Alan Grier, **2001**, *Human computers: the first pioneers of the information age*, **Endeavour** 25(1).

Setup

- Each student is assigned a position in a line.
- Each student is given an initial temperature value T_i .
- The two students at the ends of the line represent **fixed boundary conditions**.

Update rule

At each step, interior nodes update their temperature according to:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} \quad (1)$$

This means:

- you look at your two neighbours,
- take the average,
- and update your value.

Part A: Prediction

Before starting, discuss:

- What happens to a sharp temperature spike over time?
- Will the system become smoother or less smooth?

Part B: Iteration

Repeat the following for several steps:

1. Show your value to your neighbours.
2. Compute your new value.
3. Update simultaneously.

Record the evolution of the system.

Part C: Observations

Discuss:

- Where do the largest changes occur?
- What happens to temperature peaks?
- How does heat move through the system?

Part D: Interpretation

Explain in words:

- Why does the system become smoother over time?
- What role do neighbouring nodes play?
- How does this relate to diffusion?

Key Insight

Heat flow is a local process: each point evolves based on its neighbours.

Optional Extension 1: Heat Production

Introduce a constant heat source at one node.

New update rule:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} + S \quad (2)$$

where S is a constant heat input.

Tasks:

- What happens to the system over time?
 - Does it still become uniform?
 - What steady-state behaviour emerges?
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Optional Extension 2: Two-Dimensional Grid

Arrange students in a grid.

Update rule:

$$T_{i,j}^{\text{new}} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4} \quad (3)$$

Tasks:

- Observe how a hot region spreads in two dimensions.
- Compare to the one-dimensional case.
- Describe the shape of the temperature field over time.

Final Reflection

- In what sense are you “solving the heat equation”?
- Why does this method require many iterations?